

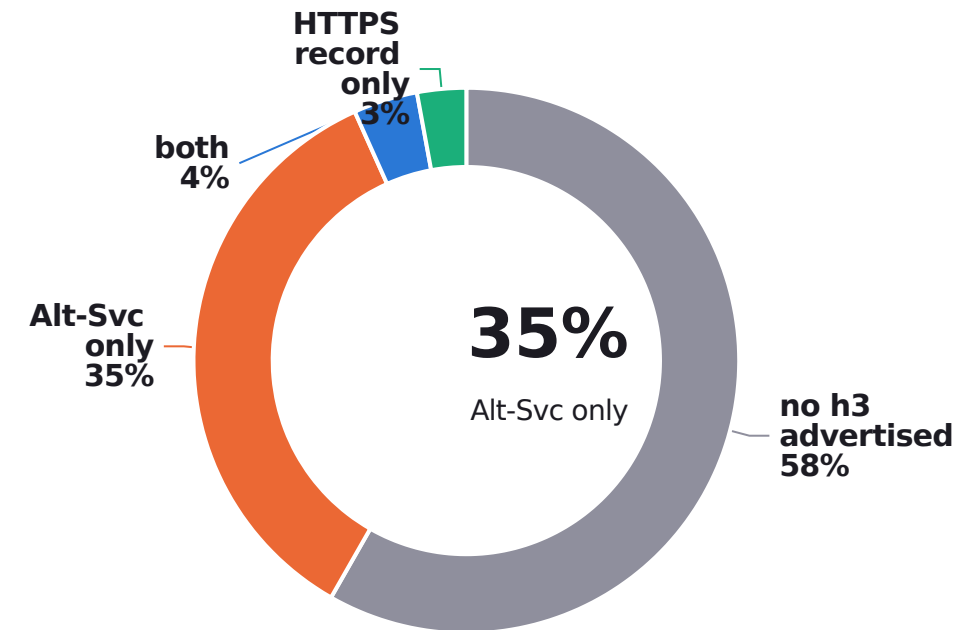
Telemetry on HTTPS RR and QUIC v2 adoption

Firefox's client-side perspective.

QUIC WG · IETF 126, Vienna · 22 July 2026 · [Max Inden](#) · Mozilla

Two upgrade paths to QUIC: Alt-Svc and HTTPS RR

- **35%** of Firefox connections learn h3 **only via Alt-Svc**, i.e. don't do HTTP/3 on the first connection ([GLAM](#)).
- Only **~7%** of connections **ever see an HTTPS record** advertising h3 ([GLAM](#)).
- HTTPS records (RFC 9460) fix this, but **few sites publish one**: **>20%** of the top-1M, and mostly one CDN's default (>70% Cloudflare) ([Dong et al., IMC 2024](#)).
- On-path actors also **strip the record** (or just its ECH parameter) to keep the SNI visible, taking the h3 hint with it ([FortiGate](#), [Cisco](#)).

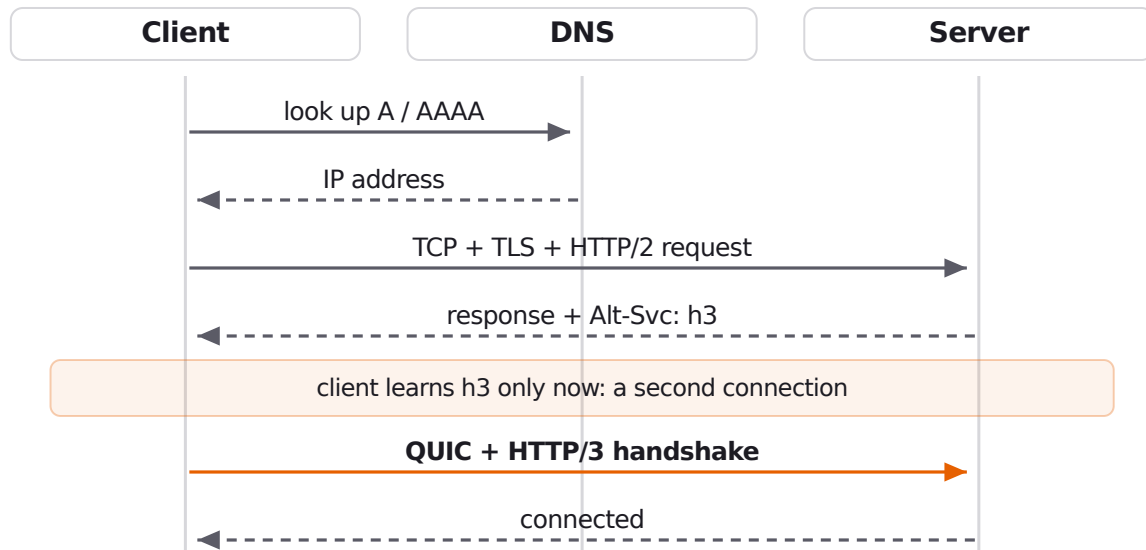


How Firefox connections learn h3. [GLAM](#).

Do we deprecate Alt-Svc, or move it into SVCB ([draft-thomson-httpbis-alt-svc](#))?

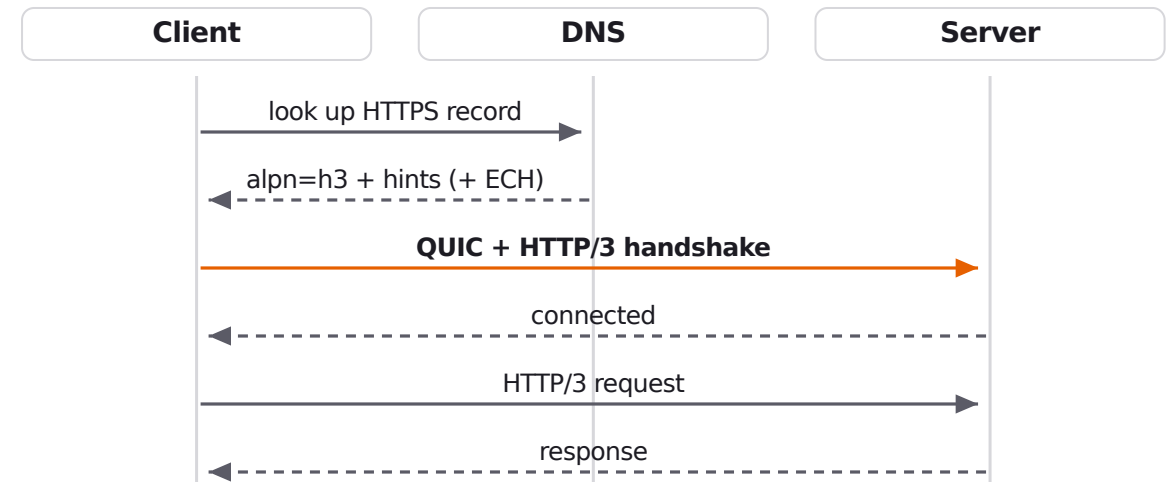
The wasted round trip

Alt-Svc only



HTTP/3 only on the second connection.

HTTPS record



HTTP/3 on the first connection.

QUIC ossification: v2 is barely deployed

- RFC 9369 is **functionally identical** to v1: only crypto salts, labels and codepoints differ (anti-ossification).
- Firefox Nightly telemetry: **v1 ~100%**, **v2 ~0%** of version-labeled QUIC handshakes ([GLAM](#)).
- Wider measurement agrees: v2 is on **< 0.003%** of QUICv1 domains ([IMC 2024](#)).
- Is adoption low because there is **no technical incentive**? v2 buys no new feature, only anti-ossification.

Who implements QUIC v2, and who runs it?

Implements v2 (10 of 14 stacks)

- [nego](#), [google/quiche](#), [ngtcp2](#)
- [msquic](#), [lsquic](#), [picoquic](#)
- [aioquic](#), [quic-go](#), [kwik](#), [HAProxy](#)

v1 only

[quinn](#), [Cloudflare quiche](#), [Meta mvfst](#), [AWS s2n-quic](#)

Deploys v2 in production (probed)

- ✓ Microsoft Outlook [outlook.office.com](#) (msquic)
- ✓ Caddy [caddyserver.com](#) (quic-go)
- ✓ LiteSpeed [litespeedtech.com](#) (lsquic)
- ✓ HAProxy [haproxy.org](#) (HAProxy)
- ✗ Google, YouTube v1
- ✗ Meta, Facebook v1
- ✗ Cloudflare v1
- ✗ Microsoft Bing v1

Implementing v2 is common, deploying it is rare. Google's stack implements v2 but its servers run v1; Microsoft runs v2 on Outlook but v1 on Bing. Probed via [savearoundtrip.com](#), 2026-07-15.

savearoundtrip.com

CHECK A DOMAIN

try: savearoundtrip.com cloudflare.com blog.mozilla.org github.com example.com

HTTPS record found: advertises h3 ✓
Browsers can negotiate HTTP/3 on the **first** connection. No round trip wasted.

mode	1 (ServiceMode)
target	. (same as owner)
alpn	☒ h3 ☐ h2
port	443
ech	not set
ttl	300s

Speaks HTTP/3 ✓
Confirmed with a live HTTP/3 handshake. The HTTPS record above advertises h3 too, so clients reach HTTP/3 on the very first connection.

Alt-Svc h3	☒ yes
HTTP/3 handshake	☒ yes
QUIC versions	☒ v1 ☒ v2
Alt-Svc	h3=":443"; ma=2592000

Share this result:

One lookup, both checks. Here quic-go.net:

- publishes an **HTTPS record** advertising **h3**, so browsers reach HTTP/3 on the **first** connection,
- and **supports QUIC v1 and v2**, confirmed by a live handshake.

Check your domain, then publish the record and turn on v2.